

# General Library Management System

A complete project proposal with requirements, ERD, participation constraints, and relational schema.

## 1. Project overview

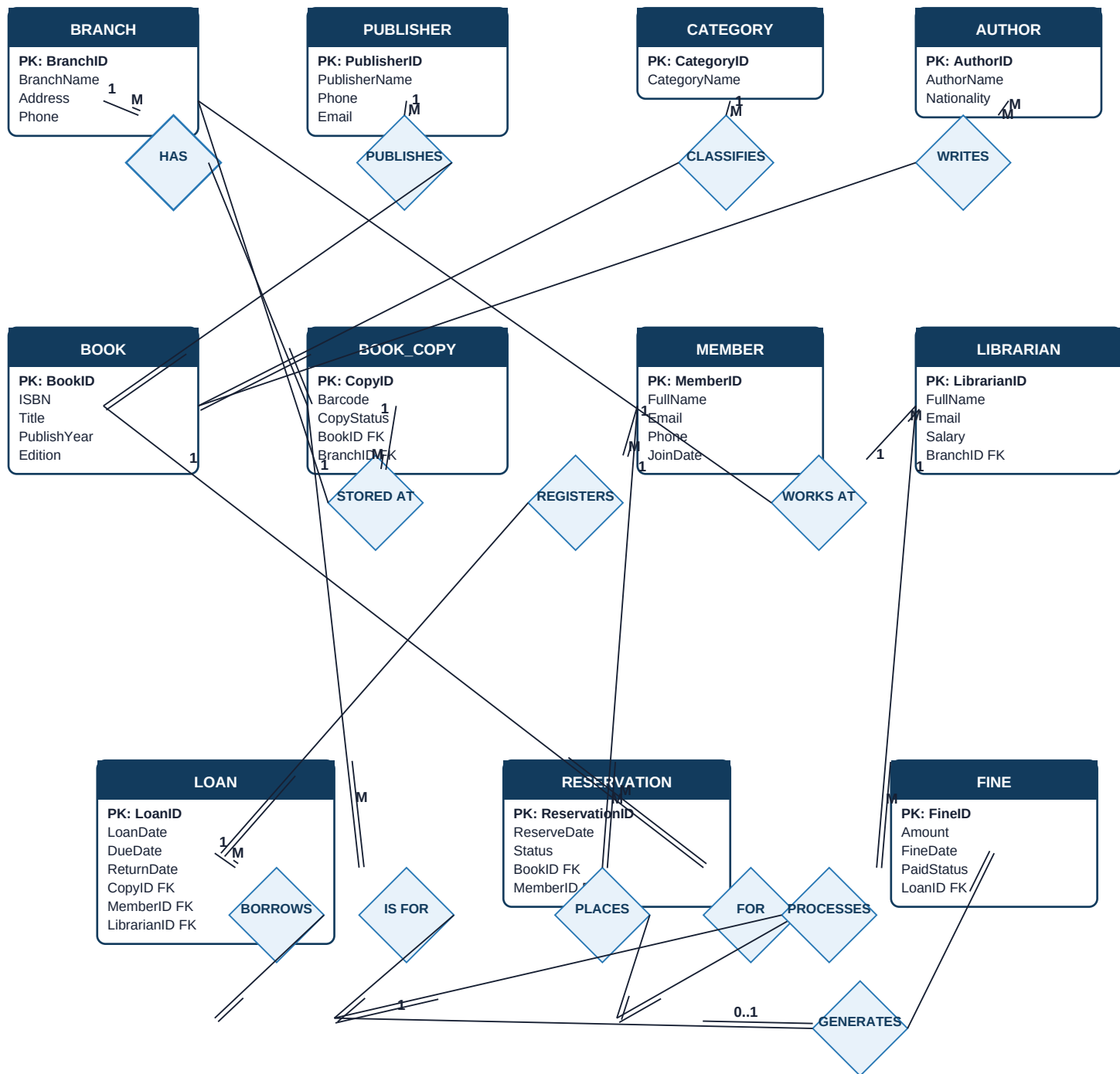
This system supports a public library with multiple branches. It manages books and copies, authors, publishers, categories, library members, loans, reservations, employees, and fines.

## 2. Business requirements

| #  | Requirement                                                                                                          |
|----|----------------------------------------------------------------------------------------------------------------------|
| 1  | The library has many branches. Each Branch has BranchID, BranchName, Address, and Phone.                             |
| 2  | The library stores books. Each Book has BookID, ISBN, Title, PublishYear, and Edition.                               |
| 3  | Each physical BookCopy belongs to one Book and is stored in one Branch. A book and branch can each have many copies. |
| 4  | Each Book is published by one Publisher. A Publisher may publish many books.                                         |
| 5  | Each Book belongs to one Category; a Category may contain many books.                                                |
| 6  | A Book can be written by many Authors, and an Author can write many Books.                                           |
| 7  | Members have MemberID, FullName, Email, Phone, and JoinDate. A member can borrow many copies over time.              |
| 8  | Each Loan records LoanID, LoanDate, DueDate, ReturnDate, and is for one BookCopy, one Member, and one Librarian.     |
| 9  | Librarians have LibrarianID, FullName, Email, Salary, and work in one Branch. A branch may employ many librarians.   |
| 10 | Members may place Reservations for books. Each Reservation records ReservationID, ReserveDate, and Status.           |
| 11 | A Loan may generate at most one Fine. A Fine stores FineID, Amount, FineDate, and PaidStatus.                        |

# ERD - General Library Management System

Single line = partial participation. A second parallel segment next to an entity = total participation.



## Reading the model

BOOK and AUTHOR are M:N through WRITES. Relationship attributes are not required in this relationship.

LOAN and RESERVATION are associative entities with their own attributes and foreign keys.

A Fine is optional for each Loan (0..1), but every Fine must refer to one Loan.

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# 3. Design explanation

## Entities (11)

BRANCH, PUBLISHER, CATEGORY, AUTHOR, BOOK, BOOK\_COPY, MEMBER, LIBRARIAN, LOAN, RESERVATION, and FINE. BOOK\_COPY represents a physical item, separating a book title from the copies held in specific branches.

## Cardinality and participation

| Relationship         | Rule                                    |
|----------------------|-----------------------------------------|
| Branch - BookCopy    | 1:M; BookCopy total, Branch partial.    |
| Publisher - Book     | 1:M; Book total, Publisher partial.     |
| Category - Book      | 1:M; Book total, Category partial.      |
| Author - Book        | M:N; both sides partial.                |
| Member - Loan        | 1:M; Loan total, Member partial.        |
| BookCopy - Loan      | 1:M; Loan total, BookCopy partial.      |
| Librarian - Loan     | 1:M; Loan total, Librarian partial.     |
| Member - Reservation | 1:M; Reservation total, Member partial. |
| Book - Reservation   | 1:M; Reservation total, Book partial.   |
| Loan - Fine          | 1:0..1; Fine total, Loan partial.       |

## 4. Relational schema

BRANCH(**BranchID**, BranchName, Address, Phone)

PUBLISHER(**PublisherID**, PublisherName, Phone, Email)

CATEGORY(**CategoryID**, CategoryName)

AUTHOR(**AuthorID**, AuthorName, Nationality)

BOOK(**BookID**, ISBN, Title, PublishYear, Edition, PublisherID FK, CategoryID FK)

BOOK\_COPY(**CopyID**, Barcode, CopyStatus, BookID FK, BranchID FK)

MEMBER(**MemberID**, FullName, Email, Phone, JoinDate)

LIBRARIAN(**LibrarianID**, FullName, Email, Salary, BranchID FK)

BOOK\_AUTHOR(**BookID FK**, **AuthorID FK**)

LOAN(**LoanID**, LoanDate, DueDate, ReturnDate, CopyID FK, MemberID FK, LibrarianID FK)

RESERVATION(**ReservationID**, ReserveDate, Status, BookID FK, MemberID FK)

FINE(**FineID**, Amount, FineDate, PaidStatus, LoanID FK UNIQUE)

Implementation note: BOOK\_AUTHOR resolves the M:N WRITES relationship. The UNIQUE constraint on FINE.LoanID enforces that a loan can generate no more than one fine.